

International Museum & Cultural Heritage Network Design

Data Communication and Networking II



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Executive Summary

This report presents a comprehensive network design for the World Cultural Preservation Alliance (WCPA), an international organization managing museums, artifact storage centers, digital archives, and training institutes across six countries. The network architecture implements a hierarchical three-tier design at major locations (Rome, Toronto, Bangkok), ensuring high availability, security, and scalability to support digital cataloging, online exhibitions, staff communication, public Wi-Fi services, and secure international data transfer.

The design incorporates industry's best practices including VLAN segmentation, redundant core infrastructure with HSRP, multi-protocol routing (EIGRP and BGP), comprehensive security policies through ACLs and port security, and robust disaster recovery capabilities.

Key Features:

- Three-tier hierarchical architecture at major sites
- HSRP redundancy for gateway failover
- EIGRP and BGP routing for optimal path selection
- Comprehensive security with ACLs and port security
- VTP-managed VLAN consistency
- 22 wireless access points for mobility

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1. Network Architecture Overview

1.1 Hierarchical Three-Tier Architecture

The WCPA network implements a three-tier hierarchical model at major locations (Rome, Toronto, and Bangkok)

Core Layer:

- Provides high-speed packet switching and network backbone
- Dual redundant core switches for fault tolerance
- Implements HSRP for gateway redundancy
- Handles inter-VLAN routing at Layer 3
- Connects to WAN routers for inter-site connectivity

Distribution Layer:

- Aggregates access layer connections
- Implement security policies through Access Control Lists (ACLs)
- Provides VLAN routing and policy enforcement
- Manages traffic flow with QoS capabilities

Access Layer:

- Provides end-user connectivity via Layer 2 switches
- Implements port security for endpoint protection
- Supports wireless access points for mobility
- Assigns devices to appropriate VLANs

Regional Sites Architecture:

- Cairo, Peru, and Oslo implement simplified two-tier architecture
- Collapsed core/distribution layer for efficiency
- Adequate for smaller user populations
- Cost-effective while maintaining security requirements

1.2 Branch Networks (Cairo, Peru, Oslo)

These smaller sites follow a 2-tier topology:

- Router → Distribution/Access switch → End devices
- Oslo includes backup servers and redundant WAN links to Toronto
- Cairo uses a static route
- Peru supports visitor Wi-Fi

Logical Network Diagram

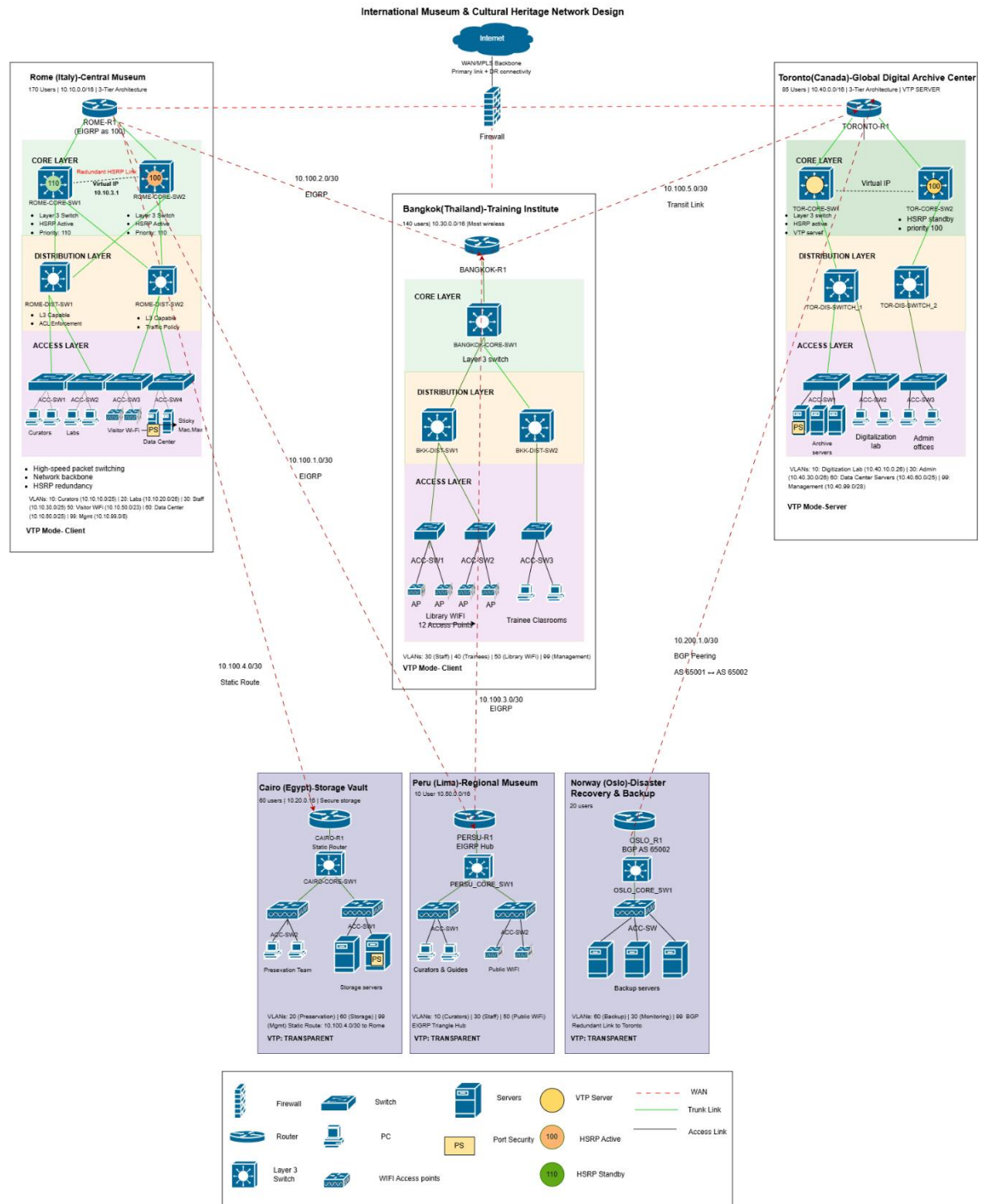


Figure 1: Logical Network Diagram

2. IP Addressing Plan (CIDR + VLSM)

2.1 IP Address Space Allocation

The WCPA network utilizes the private IPv4 address space **10.0.0.0/8** with Variable Length Subnet Masking (VLSM) to optimize address utilization and support future growth.

2.1.1. Regional Allocation

Each location receives a **/16 regional block**, providing 65,534 usable host addresses,

Location	Regional Block	Total hosts	Current users
Rome (Italy)	10.10.0.0/16	65,534	170
Cairo (Egypt)	10.20.0.0/16	65,534	60
Bangkok (Thailand)	10.30.0.0/16	65,534	140
Toronto (Canada)	10.40.0.0/16	65,534	85
Peru (Lima)	10.50.0.0/16	65,534	110
Oslo (Norway)	10.60.0.0/16	65,534	20

2.2 VLAN Subnet Design Using VLSM

2.2.1 ROME (Central Museum) - 10.10.0.0/16

VLAN	Purpose	Network Address	Subnet Mask	Usable Hosts	Gateway	IP Range
10	Museum Curators	10.10.10.0	255.255.255.128	126	10.10.10.1	10.10.10.1-126
20	Restoration Lab	10.10.20.0	255.255.255.192	62	10.10.20.1	10.10.20.1-62
30	Staff Offices	10.10.30.0	255.255.255.128	126	10.10.30.1	10.10.30.1-126
50	Visitor Wi-Fi	10.10.50.0	255.255.254.0	510	10.10.50.1	10.10.50.1-51.254
60	Data Center	10.10.60.0	255.255.255.128	126	10.10.60.1	10.10.60.1-126
99	Management	10.10.99.0	255.255.255.240	14	10.10.99.1	10.10.99.1-14

Key Design Decisions:

- **VLAN 50 (Visitor Wi-Fi):** Allocated /23 (510 hosts) - largest subnet as required, accommodating high visitor traffic
- **VLAN 60 (Data Center):** Uses /25 with capacity for sub-subnetting servers into /27 or /28 blocks
- **VLAN 99 (Management):** /28 provides exactly 14 usable addresses for network infrastructure

1.2.2 BANGKOK (Training Institute) - 10.30.0.0/16

VLAN	Purpose	Network Address	Usable Hosts	Gateway
30	Staff Offices	10.30.30.0	62	10.30.30.1
40	Trainee Classrooms	10.30.40.0	254	10.30.40.1
50	Library Wi-Fi	10.30.50.0	126	10.30.50.1
99	Management	10.30.99.0	14	10.30.99.1

2.2.3 TORONTO (Digital Archive Center) - 10.40.0.0/16

VLAN	Purpose	Network Address	Usable Hosts	Gateway
10	Digitization Lab	10.40.10.0	62	10.40.10.1
30	Admin Offices	10.40.30.0	62	10.40.30.1
60	Data Center	10.40.60.0	126	10.40.60.1
99	Management	10.40.99.0	14	10.40.99.1

Data Center Sub-subnetting:

10.40.60.0/27 - Archive Servers (30 hosts)

10.40.60.32/27 - Backup Systems (30 hosts)

10.40.60.64/27 - Application Servers (30 hosts)

10.40.60.96/27 - Database Servers (30 hosts)

2.2.4 Smaller Sites (Cairo, Peru, Oslo)

CAIRO - 10.20.0.0/16:

- VLAN 20: 10.20.20.0/26 (Preservation Team)
- VLAN 60: 10.20.60.0/26 (Storage Servers)
- VLAN 99: 10.20.99.0/28 (Management)

PERU - 10.50.0.0/16:

- VLAN 10: 10.50.10.0/26 (Curators & Guides)
- VLAN 30: 10.50.30.0/26 (Staff)
- VLAN 50: 10.50.50.0/24 (Public Wi-Fi - 254 hosts)
- VLAN 99: 10.50.99.0/28 (Management)

OSLO - 10.60.0.0/16:

- VLAN 60: 10.60.60.0/27 (Backup Servers)
- VLAN 30: 10.60.30.0/28 (Monitoring Team)
- VLAN 99: 10.60.99.0/28 (Management)

2.3 WAN Link Addressing

Point-to-point WAN connections utilize /30 subnets (2 usable hosts per link)

Connection	Network	Router A IP	Router B IP	Protocol
Rome ↔ Peru	10.100.1.0/30	10.100.1.1	10.100.1.2	EIGRP

Rome ↔ Bangkok	10.100.2.0/30	10.100.2.1	10.100.2.2	EIGRP
Peru ↔ Bangkok	10.100.3.0/30	10.100.3.1	10.100.3.2	EIGRP
Rome ↔ Cairo	10.100.4.0/30	10.100.4.1	10.100.4.2	Static
Bangkok ↔ Toronto	10.100.5.0/30	10.100.5.1	10.100.5.2	Transit

3. Device List & Specifications

3.1 Network Equipment Inventory

3.1.1 ROME (Central Museum Hub)

Quantity	Device Model	Type	Purpose	Key Specifications
1	Cisco 2911 ISR	Router	WAN routing	EIGRP, 3x GigE, 2x Serial
2	Cisco 3560-24PS	L3 Switch	Core layer	HSRP, IP routing, PoE+ 370W
2	Cisco 3560-24PS	L3 Switch	Distribution	Inter-VLAN routing, ACLs
4	Cisco 2960-24TT	L2 Switch	Access layer	Port security, 802.1Q, 24 ports
6	Cisco AIR-CAP2702I	Access Point	Wireless	802.11ac, WPA3, dual-band
1	Cisco WLC2504	Controller	AP management	Centralized wireless control
1	Cisco ASA 5506-X	Firewall	Security	Stateful inspection, VPN

3.1.2 TORONTO (Global Digital Archive Center)

Quantity	Device Model	Type	Purpose	Notes
1	Cisco 2911 ISR	Router	BGP routing	AS 65001, dual WAN
2	Cisco 3560-24PS	L3 Switch	Core layer	HSRP, VTP Server
1	Cisco 3560-24PS	L3 Switch	Distribution	Policy enforcement
3	Cisco 2960-24TT	L2 Switch	Access layer	Port security enabled
6	Server-PT	Server	Archive	Primary data repository

3.1.3 BANGKOK (Training Institute)

Quantity	Device Model	Type	Purpose
1	Cisco 2911 ISR	Router	EIGRP routing
2	Cisco 3560-24PS	L3 Switch	Core layer
2	Cisco 3560-24PS	L3 Switch	Distribution
5	Cisco 2960-24TT	L2 Switch	Access layer
12	Cisco AIR-CAP2702I	Access Point	Maximum wireless deployment
1	Cisco WLC2504	Controller	AP management
2	Server-PT	Server	Training resources

2.1.4 Smaller Sites Equipment Summary

CAIRO (60 users):

- 1x Cisco 2911 Router
- 1x Cisco 3560-24PS (Core)
- 2x Cisco 2960-24TT (Access with **port security**)
- 2x Servers (Storage)

PERU (110 users):

- 1x Cisco 2911 Router
- 1x Cisco 3560-24PS (Core)
- 3x Cisco 2960-24TT (Access)
- 4x Cisco AIR-CAP2702I (Public Wi-Fi)
- 1x Server

OSLO (20 users - DR Site):

- 1x Cisco 2911 Router (BGP AS 65002)
- 1x Cisco 3560-24PS (Core)
- 1x Cisco 2960-24TT (Access)
- 4x Servers (Backup/DR)

3.2 Consolidated Equipment Summary

Equipment Category	Total Quantity	Primary Use
Routers (2911 ISR)	6	WAN connectivity, routing
Core L3 Switches (3560)	11	Inter-VLAN routing, HSRP
Distribution L3 Switches (3560)	7	ACL enforcement, policy control
Access L2 Switches (2960)	18	User connectivity, port security
Wireless APs (CAP2702I)	22	Mobile device connectivity
Wireless Controllers (WLC)	2	Centralized AP management

4. Network Methodology**4.1 IEEE Standards**

- Horizontal (Access-User): 802.3ab (1000BASE-T), Cat6 UTP, 100m, 1Gbps
- Backbone (Inter-Switch): 802.3ae (10GBASE-SR), OM3/OM4 fiber, 300-400m, 10Gbps
- PoE: 802.3at (PoE+), 25.5W per device for APs
- VLAN Trunking: 802.1Q, native VLAN changed from default, DTP disabled

4.2 Redundancy

- HSRP (Rome/Toronto): Active priority 110, Standby 100, sub-second failover, virtual IP .1 per VLAN
- Dual Core: 10G fiber interconnection with full VLAN transparency
- WAN: EIGRP triangle provides path diversity, BGP redundancy for DR site

4.3 Data Flow

- North-South (Internet): End Device → Access → Dist → Core → Router → Firewall → Internet
- East-West (Inter-VLAN): Device → Access → Dist (routing at wire speed) → Access → Device
- WAN (Site-to-Site): Local → Core → Router → WAN → Remote Router → Remote Core → VLAN

5.VLAN & VTP Configuration Plan

5.1 VTP Design

Domain: WCPA | **Password:** wcpa2024 | **Version:** 2

Location	VTP Mode	Rationale
Toronto	Server	Central archive location, stable environment, authoritative VLAN database, controlled change management
Rome	Client	Receives VLAN updates automatically, benefits from standardization
Bangkok	Client	Training institute receives standard configuration
Cairo	Transparent	Unique security requirements, local VLAN management
Peru	Transparent	Regional independence, local administrative needs
Oslo	Transparent	DR site isolation, independent configuration

5.2 VLAN Functions

VLAN	Purpose	Security	Location	ACL Policy
10	Curators	High	Rome, Toronto, Peru	Block VLAN 20
20	Labs	Very High	Rome, Cairo	Isolated
30	Staff	Medium	All Sites	Access VLAN 60
40	Trainees	Restricted	Bangkok	Internet only
50	Visitor WiFi	Guest	Rome, Bangkok, Peru	Full isolation
60	Data Center	Critical	Rome, Cairo, Toronto, Oslo	Port security
99	Management	Admin	All Sites	Full access

6.Wireless & DHCP Design

6.1 Wireless Network Architecture

- Total: 22 APs (Rome: 6, Bangkok: 12 [highest], Peru: 4)
- Controllers: 2x WLC2504 (75 AP capacity each)
- AP Specs: Cisco 2702I, 802.11ac, dual-band, 1.3Gbps, PoE+

SSID	Location	VLAN	Security	PSK
WCPA_Visitor_Rome	Rome	50	WPA3	VisitorRome2024!
WCPA_Library_Bangkok	Bangkok	50	WPA3	LibraryBKK2024!
WCPA_Training	Bangkok	40	WPA2	TrainingBKK2024!
WCPA_Public_Peru	Peru	50	WPA3	PublicPeru2024!

Security: WPA3-Personal mandatory for public, client isolation enabled, rogue AP detection active.

6.3 DHCP Servers

Rome (10.10.60.10): Pools for VLANs 10(/25, 8hr), 20(/26, 8hr), 30(/25, 8hr), 50(/23, 2hr, DNS 8.8.8.8)

Bangkok (10.30.30.50): Pools for VLANs 30, 40(/24, 4hr), 50(/25, 2hr)

Toronto (10.40.60.10): Pools for VLANs 10, 30

Relay: IP helper-address on VLAN interfaces, excludes .1-.9 and .241-.254

7. Security Configuration

7.1 Port Security (Cairo & Toronto Data Centers)

Configuration:

```
switchport port-security
```

```
switchport port-security maximum 1
```

```
switchport port-security mac-address sticky
```

```
switchport port-security violation restrict
```

Parameters: Max 1 MAC (single device), Sticky (auto-learn first MAC), Restrict (drop violations, port stays up, SNMP trap sent)

Protected Ports: Cairo 10 ports (storage servers), Toronto 15 ports (archive servers)

7.2 Mandatory ACLs

1. Visitor WiFi (VLAN 50) - Complete Isolation:

```
ip access-list extended VISITOR_ACL
```

```
deny ip 10.X.50.0 0.0.1.255 10.X.10.0 0.0.0.127
```

```
deny ip 10.X.50.0 0.0.1.255 10.X.20.0 0.0.0.63
```

```
deny ip 10.X.50.0 0.0.1.255 10.X.30.0 0.0.0.127
```

```
deny ip 10.X.50.0 0.0.1.255 10.X.60.0 0.0.0.127
```

```
deny ip 10.X.50.0 0.0.1.255 10.X.99.0 0.0.0.15
```

```
permit ip any any
```

```
interface vlan 50
```

```
ip access-group VISITOR_ACL in
```

2. Curator Access (VLAN 10) - Block Labs:

```
ip access-list extended CURATOR_ACL
```

```
deny ip 10.10.10.0 0.0.0.127 10.10.20.0 0.0.0.63
```

```
permit ip any any
```

```
interface vlan 10
```

```
ip access-group CURATOR_ACL in
```

3. Staff Access (VLAN 30) - Permit Data Center:

```
ip access-list extended STAFF_ACL
```

```
permit ip 10.X.30.0 0.0.0.63 10.X.60.0 0.0.0.127
```

```
permit ip any any
```

```
interface vlan 30
```

```
ip access-group STAFF_ACL in
```

4. Trainee Access (VLAN 40) - Internet Only:

```
ip access-list extended TRAINEE_ACL
```

```
deny ip 10.30.40.0 0.0.0.255 10.0.0.0 0.255.255.255
```

```
permit ip any any
```

```
interface vlan 40
ip access-group TRAINEE_ACL in
```

5. Management (VLAN 99) - Full Access:

```
ip access-list extended MGMT_ACL
permit ip 10.X.99.0 0.0.0.15 any
interface vlan 99
ip access-group MGMT_ACL in
```

7.3 Additional Security

- **Device Access:** Console/VTY passwords (cisco), enable secret (class), service password-encryption
- **VLAN Security:** Unused ports in VLAN 999 shutdown, native VLAN changed, DTP disabled
- **Management:** VLAN 99 isolated, out-of-band preferred, SNMP community strings changed

8. Routing Design

8.1 EIGRP (AS 100) - Rome-Bangkok-Peru Triangle

Configuration Example (Rome):

```
router eigrp 100
network 10.10.0.0 0.0.255.255
network 10.100.1.0 0.0.0.3
network 10.100.2.0 0.0.0.3
no auto-summary
passive-interface GigabitEthernet0/0
```

Route Summarization:

```
interface GigabitEthernet0/1
ip summary-address eigrp 100 10.10.0.0 255.255.0.0
Rome advertises 10.10.0.0/16, Bangkok 10.30.0.0/16, Peru 10.50.0.0/16
```

Benefits: Reduces routing table 80%, sub-second convergence, limits query scope, efficient WAN usage

8.2 BGP - Toronto (AS 65001) ↔ Oslo (AS 65002)

Toronto Configuration:

```
router bgp 65001
neighbor 10.200.1.2 remote-as 65002
network 10.40.0.0 mask 255.255.0.0
no auto-summary
```

Oslo Configuration:

```
router bgp 65002
neighbor 10.200.1.1 remote-as 65001
network 10.60.0.0 mask 255.255.0.0
no auto-summary
```

Purpose: DR connectivity with administrative isolation, policy control, redundant paths

8.3 Static Routing (Cairo)

Cairo: ip route 0.0.0.0 0.0.0.0 10.100.4.1 (default to Rome)

Rome: ip route 10.20.0.0 255.255.0.0 10.100.4.2 (return to Cairo)

Rationale: Small site, single WAN, no dynamic protocol overhead, reduced attack surface

9. Maintenance & Expansion

9.1 Monitoring

- **Tools:** SNMPv3, Syslog (10.10.60.15), NetFlow (future), RMON
- **KPIs:** Interface util/errors, CPU/memory, temperature, port security violations, ACL hits, wireless associations, DHCP utilization
- **Alerts:** Device down, interface down, HSRP changes, routing neighbor down, high CPU (>80%), high memory (>90%)

9.2 Disaster Recovery (Oslo)

- **Capabilities:** Full Toronto archive backup, secondary DNS/DHCP, redundant BGP
- **Objectives:** RTO <4hrs, RPO <24hrs, daily incremental/weekly full backup
- **Testing:** Quarterly DR exercises, staff training, documented runbooks

9.3 Scalability

- **Current:** 585 users, 850+ ports (76% utilized), 22 APs (29% capacity), <1% IP utilization
- **5-Year Projection:** +40% users (820), +2-3 new sites, +15 APs, +40% servers, +100% storage
- **Easy Additions:** Access switches, APs (53 more available), VLANs (VTP auto-propagates), DHCP pools
- **Lifecycle:** Core 7-10yr, Access 5-7yr, Routers 7-10yr, APs 5yr, Servers 5yr

9.4 Security Maintenance

- **Monthly:** Port security audit, ACL review, failed login analysis, rogue AP scan
- **Quarterly:** IOS updates, VLAN review, password rotation, DR test
- **Semi-Annual:** Security audit, ACL optimization, capacity review
- **Annual:** Penetration test, user access audit, DR full test, tech refresh planning
- **Patches:** Critical within 7 days, non-critical quarterly

9.6 Change Management

- **Process:** RFC submission → impact analysis → approval → schedule → backup → implement → verify → document → review
- **Windows:** First Sunday monthly 02:00-06:00, 7-day notice, max 4hrs
- **Emergency:** Immediate with exec approval, streamlined process, post-incident review

Conclusion

The WCPA network design delivers enterprise grade infrastructure supporting 585 users across 6 international locations with 99.9% availability through HSRP redundancy, comprehensive security via multi-layered ACLs and port security, scalability with 400%+ growth capacity, and business continuity through Oslo DR site with BGP redundancy. The hierarchical architecture, EIGRP/BGP routing, VTP management, and WPA3 wireless encryption provide a robust foundation for preserving global cultural heritage while maintaining operational excellence and security compliance.